REJUVENATION STRATEGY OF THE DYING SPRINGS AND NAULAS IN DHARGAD WATERSHED (JAIGAN VALLEY) DISTRICT BAGESHWAR, UTTARAKHAND

Phase-I Rejuvenation of Springs & Naulas in Jatha & Pass Villages

SPRINGSHED DELINEATION & AQUIFER RECHARGE

Target Settlements: Jatha & Pass Villages

Watershed: Dhargad Sub-Catchment, Jaigan Valley

Elevation: 1,150 m to 1,920 m above MSL

Integrated Geospatial Hydrology & Field Micro-Plans

Prof. J.S. Rawat

Er. Varun Rawat

Dr. N.C. Pant

Sponsored by:

AMAN

Eastern Pokherkhali, Almora, Uttarakhand, India

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1.0 EXECUTIVE OVERVIEW & STUDY AREA

The Dhargad Watershed is an integral tributary catchment within the Jaigan River basin in District Bageshwar, Uttarakhand. Perennial springs (Dharas) and traditional subterranean recharge step-wells (Naulas) have formed the lifeline of drinking water and terraced agriculture for generations in Jatha and Pass villages. Over recent decades, intensified monsoonal runoff, structural fracture dislocation, and changes in vegetative land cover have severely reduced lean-season spring discharge, precipitating acute seasonal water scarcity.

1.1 Geospatial Hydrogeology & Recharge Zone Delineation

Using high-resolution Digital Elevation Models (DEMs), multi-spectral satellite imagery, and on-ground geologic fracture tracing, the recharge zones contributing to the principal dying springs of Jatha and Pass were precisely mapped. The study area extends across 1.42 km² of mountainous terrain, characterized by folded quartzite, phyllite bedrock, and thin colluvial slope mantles.

Table 1: Planned Civil & Biological Treatments in Dhargad Watershed

Treatment Measure	Unit Target	Target Zone	Hydrological Function
Infiltration Holes (5/m²)	3,200,000 units	Degraded ridge slopes	Direct fracture infiltration & runoff arrest
Infiltration Trenches (1.0x0.5x0.3m)	11,400 trenches	Barren & scrub slopes	Contour detention of monsoon sheet wash
Pine Needle Bio-Plugs	45 units	Upper 1st-order rills	Sediment trapping and runoff velocity dissipa
Loose Stone Gully Plugs	72 units	2nd-order gullies	Preventing incisive channel downcutting
Broad-Leaved Afforestation	18,500 saplings	Catchment commons	Quercus leucotrichophora (Oak) & Alnus (Ut

2.0 ACTION PLAN & IMPLEMENTATION ROADMAP

The action plan emphasizes participatory springshed management by establishing Village Water User Committees (Gram Jal Samitis) in Jatha and Pass. Community labor combined with micro-watershed engineering ensures low-cost, high-durability recharge infrastructure that stabilizes local potable water access year-round.

INVESTIGATION TEAM & EDITORIAL SIGN-OFF

Prof. J.S. Rawat
Senior Hydrogeologist & Geoscientist
Former Professor & Head, Kumaun University

Er. Varun Rawat
Geospatial Engineer & Founder
GEOVERSED Geoscience Intelligence

Dr. N.C. Pant
Co-Principal Investigator
Watershed & Environmental Scientist

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